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ETRS89 in the EPSG Dataset

EPSG
GEODETIC PARAMETER DATASET
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About

This Guidance Note explains how ETRS89 regional and national realizations are available in the EPSG Dataset, in particular related to naming conventions and use of the datum ensemble. This caters to highest-accuracy users who require exact definition of coordinate epoch and realization as well as regular users who can work within the accuracy of the ETRS89 ensemble.

The Guidance Note is aimed at users of the EPSG Dataset, in particular software developers, positioning service providers and staff in operating companies.

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ETRS89 in the EPSG Dataset

Revision history

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Introduction

The EPSG Geodetic Parameter Dataset, abbreviated to the **EPSG Dataset**, is a repository of parameters required to:

- define a *coordinate reference system* (CRS) which ensures that coordinates describe position unambiguously;
- define transformations and conversions that allow coordinates to be changed from one CRS to another CRS. Transformations and conversions are collectively called *coordinate operations*.

The EPSG Dataset is maintained by the IOGP. It conforms to ISO 19111:2019 – *Referencing by coordinates*. It is distributed in three ways:

- the **EPSG Registry**, in full the *EPSG Geodetic Parameter Registry*, a web-based delivery platform with a graphic user interface (GUI) and an application programming interface (API). From this repository descriptions of CRSs and transformations may be output in well-known text (WKT) conformant to ISO 19162 or as GML documents;
- the **EPSG Database**, in full *the EPSG Geodetic Parameter Database*, an MS Access relational database with content derived from the online registry;
- in a relational data model as **SQL scripts** which enable a user to create an Oracle, MySQL, PostgreSQL or other relational database and populate that database with the EPSG Dataset.

The terms of use of the EPSG Dataset are available here.

IOGP Reports 373-01-1 through 373-07-7 form a multi-part document for users of the EPSG Dataset:

- IOGP Report 373-07-1 (EPSG Guidance Note 7-1) – *Understanding the EPSG Geodetic Parameter Dataset*, sets out detailed information about the Dataset content and its maintenance.
- IOGP Report 373-07-2 (EPSG Guidance Note 7-2) – *Formulas* provides a detailed explanation of formulas necessary for executing coordinate conversions and transformations using the coordinate operation methods supported in the EPSG Dataset. Geodetic parameters in the EPSG Dataset are consistent with these formulas.
- IOGP Report 373-07-3 (EPSG Guidance Note 7-3) – *EPSG Registry API User Guide*, is primarily intended to assist computer application developers who wish to use the RESTful API of the EPSG Registry to query and retrieve entities and attributes from the Dataset.
- IOGP Report 373-07-4 (EPSG Guidance Note 7-4) – *EPSG Database and SQL Script User Guide*, provides guidance for users of the EPSG Database and SQL scripts which may be obtained from the EPSG Registry's *Download Dataset* page.
- IOGP Report 373-07-5 (EPSG Guidance Note 7-5) – *EPSG null and copy transformations to WGS 84*, explains aspects of the policies for the inclusion in the EPSG Dataset of datum ensembles and so-called null coordinate transformations to WGS 84, including limitations in the description of WGS 84 through code EPSG:4326.
- IOGP Report 373-07-6 (EPSG Guidance Note 7-6) – *EPSG policies and procedures for EPSG Dataset data management,* documents strategy for populating the EPSG Dataset.
- IOGP Report 373-07-7 (EPSG Guidance Note 7-7) - *ETRS89 in the EPSG Dataset* (this document), describes EPSG dataset records for the European Terrestrial Reference System of 1989.

The complete texts may be found on the EPSG Registry web site under Support Documentation.

1. ETRS89

The European Terrestrial Reference System of 1989, ETRS89, is the regional reference system for Europe. Using modern geodetic terminology, it is a terrestrial reference system (hence E*TRS*). ETRS89 is defined as being fixed to the stable part of the EurAsian tectonic plate and aligned with the International Terrestrial Reference System (ITRS) at epoch 1989.0 [EUREF, 2024]. ETRS89 has been realized through the European Terrestrial Reference Frame, ETRF, with a new ETRFyy being defined with the publication of each realization of the International Terrestrial Reference Frame, ITRFyy, except for ITRF2008. There are currently 12 pan-European realizations of ETRS89: ETRF89 through to ETRF2020.

At the time ETRS89 was created, 'fixed to the stable part of the EurAsian tectonic plate' implied that ETRS89 and ETRFyy were static. With the passage of time and improvements in positioning technology, particularly observations at continuously operating GNSS stations, changes in coordinates of locations on the crust of the Earth are now measurable, and ETRS89 and ETRFyy are now considered to be dynamic - coordinates change with time (horizontally and/or vertically).

In addition to the pan-European ETRFyy realizations of ETRS89, most countries have their own national realization. National realizations of ETRS89 are typically defined through the coordinates of a network of national and some surrounding regional EUREF Permanent Network (EPN) GNSS stations. The national network is constrained to ITRS or IGS (ITRFyy or IGSyy) coordinates at a specified epoch and then aligned to a specified ETRFyy. An example of this (from Great Britain) is "ITRF97 at epoch 2009.756 aligned to ETRF97 following the EUREF conventions". These national realizations are static snapshots of ETRS89. Because each national realization is a separate densification of a differing set of EPN stations and different ETRFs have been used for alignment, there are small - centimetre level - differences between different national realizations. Tectonic activity in or near southern Europe and glacial isostatic adjustment in northern Europe cause further changes in coordinates with time. For many practical purposes these small differences are insignificant, but for geodesy and other high-accuracy purposes such as cross-border civil engineering projects the differences are significant.

The EPSG Dataset documents pan-European and national realizations of ETRS89 and provides paths for merging spatial datasets referenced to these realizations. IOGP received comments that it was not easy to find whether Coordinate Reference Systems (CRSs) were based on national realizations of ETRS89. In 2025, IOGP presented proposals to EUREF workshops and symposia [EUREF, 2025] for updates to the EPSG Dataset to make this task easier for Dataset users. These were implemented in the latter part of 2025. This Guidance Note discusses how realizations of ETRS89 documented in the EPSG Dataset [IOGP, 2026] may now be identified and related to each other.

The remainder of this document uses terminology used in ISO 19111:2019 and the EPSG Dataset which has its origins in classical geodesy; in the EPSG Dataset, "ETRS89" is the name of a coordinate reference system (CRS) based on a datum ensemble. The datum ensemble concept is discussed in the following section.

2. The ETRS89 datum ensemble

A datum ensemble is an artificial data modelling construct described in ISO 19111 [ISO, 2019]. It documents a group of datums which for lower-accuracy purposes can be considered to be insignificantly different from each other. The merging of data referenced to the ETRS89 datum ensemble is discussed in section 6.

The accuracy at which the assumption of no significant difference applies is given by the *datum ensemble accuracy* attribute. For the ETRS89 datum ensemble, this has been recorded in the EPSG Dataset as 0.1 m.

Use of a datum ensemble comes with a health warning. Spatial data related to it lacks any information about which specific member of the ensemble the data is referenced to. The data therefore becomes 'low spatial threshold use'. In the case of the ETRS89 ensemble, the data accuracy is no better than the ensemble accuracy of 0.1 metre. Spatial data requiring highest accuracy should always be referenced to a specific realization.

In the EPSG Dataset, prior to October 2025 the members of the ETRS89 ensemble were the pan-European ETRFyy realizations. From October 2025, national realizations of ETRS89 have also been included.

Geocentric, geographic 3D and geographic 2D CRSs based directly on the ETRS89 datum ensemble have EPSG CRS codes 4936, 4937 and 4258, respectively. Numerous projected CRSs are based on EPSG:4258. Despite the inclusion of national realizations of ETRS89, the EPSG Dataset includes ensemble-based CRSs and may have transformations associated with them. This is particularly the case in offshore areas where positioning may be through services other than national CORS systems. In such cases use of these ensemble-based CRSs may duplicate use of national realizations. To the 0.1 m accuracy of the ensemble, the ensembled-based and national CRSs may be related through a null transformation, discussed further in section 6.5. Care in data management is required.

3. EPSG naming convention for national ETRS89 realizations

There has been no pan-European standard for naming of national realizations. Some countries include ETRS or ETRF in their nationally recognized name, for example “ETRF2000-PL” for Poland. But for the uninitiated unfamiliar with the geodetic terminology of reference system and reference frame, ETRF2000 does not have any obvious connection to ETRS89. Other countries use local names, for example “RGF93” in France or “SWEREF 99” in Sweden. These names give little indication that the CRS is a realization of ETRS89.

To make the names of national realizations more obviously related to ETRS89, from October 2025 EPSG has adopted the following standard syntax for CRS names:

- ETRS89-ABC [local name]

where “ABC” is the ISO 3166 three-character country code and “local name” is substituted with the locally used name (or an English transliteration of this). Some examples are:

- ETRS89-SWE [SWEREF 99]
- ETRS89-POL [ETRF2000-PL]
- ETRS89-FRA [RGF93]

The name "ETRS89" without amplification is retained as the name for geocentric, geographic 3D and geographic 2D CRSs based on the ETRS89 datum ensemble.

In the EPSG Dataset, the name of the datum upon which the national CRS is based is usually the locally recognized name (or an English transliteration of this), with the EPSG conventional CRS name syntax given as an alias for the datum name.

4. Projected CRSs

The INSPIRE[1] Data Specification on Coordinate Reference Systems [INSPIRE, 2007] recommends:

- The UTM (Universal Transverse Mercator) projection for conformal pan-European mapping at scales larger than 1:500,000.
- The ETRS89-LCC (Lambert Conformal Conic) projection for conformal pan-European mapping at scales smaller than or equal to 1:500,000.
- The ETRS89-LAEA (Lambert Azimuthal Equal Area) projection for pan-European spatial analysis and reporting where true area representation is required.

In the EPSG Dataset, these projections are applied to the ensemble-based ETRS89 geographic 2D CRS, EPSG:4258. They may also be in use with national realizations of ETRS89; when this is the case, care in spatial data management is required as in effect there are duplicate EPSG records for the same projected CRS.

4.1. UTM

The EPSG codes for projected CRSs based on the ETRS89 ensemble for UTM zones 28 to 37 are 25828 to 25837 for CRSs in which the coordinate axis order is easting, northing and 3040 to 3049 for CRSs in which the coordinate axis order is northing, easting. Separate records and codes coexist for UTM zones for projected CRSs based on national realizations of ETRS89, e.g., code 11145 for ETRS89-ESP [REGENTE] / UTM zone 30N in Spain, and code 11020 for ETRS89-NOR [EUREF89] / UTM zone 30N in Norway.

A projected CRS applied to national realizations of ETRS89 may have defining parameters identical to those of UTM but may carry a local name that is not obviously UTM. For example, in Finland the TMFIN35 projection has identical map projection parameters to UTM zone 35N. When this is the case, the EPSG policy is to add an EPSG alias containing "... UTM zone xxN" in the projected CRS record.

A text search in the EPSG Registry at epsg.org for "ETRS89 / UTM zone 35N" will return both the ensemble based EPSG:25835 and EPSG:3047 projected CRSs as well as any zone 35N projected CRSs based on national realizations such as "ETRS89-FIN [EUREF-FIN] / TMFIN35" in Finland and "ETRS89-NOR [EUREF89] / UTM zone 35N" in Norway.

4.2. ETRS89-LCC and ETRS89-LAEA

In the EPSG Dataset, the extent assigned to pan-European realizations of ETRS89 (the ETRS89 datum ensemble and ETRFyy) is limited to those countries on the EurAsian tectonic plate where ETRS89 is in use. There are some areas of the stable part of the tectonic plate that are excluded from this extent because ETRS89 is not in practical use, even though there are GNSS stations of the EPN network in the country, for example in Belarus.

[1] Infrastructure for Spatial Information in Europe.

At the request of the European Union Joint Research Council, in the EPSG Dataset the extents for the ETRS89-LCC and ETRS89-LAEA projections and projected CRSs (EPSG:3034 and EPSG:3035) have been extended to include several European areas not on the EurAsian tectonic plate. This was done for example for Cyprus, Iceland and the Atlantic islands of Portugal and Spain, where other geodetic datums are in use. These projected CRSs carry the name "ETRS89-extended" (e.g., "ETRS89-extended / LCC Europe" for EPSG:3034).

5. How to find national realizations of ETRS89

Prior to October 2025, other than for a very small number of realizations where the national authority included ETRS89 or ETRF in the name, the only indication whether data in the EPSG Dataset represented a national realization of ETRS89 was through free-text remarks in geodetic datum and geodetic CRS records.

Since October 2025, three other routes are available:

i) The name of the national realization CRS uses the standard syntax, beginning with "ETRS89-" and ending with a closing square bracket.

   A text search for 'ETRS89 / UTM zone 35N' will return both the ensemble based EPSG:25835 and EPSG:3047 projected CRSs as well as any zone 35N projected CRSs based on national realizations such as "ETRS89-FIN [EUREF-FIN] / TMFIN35" in Finland and "ETRS89-NOR [EUREF89] / UTM zone 35N" in Norway.

ii) The CRS is based on a geodetic datum which has its conventional reference system attribute populated as code 1025.

   This may be decoded through the ConventionalRS table as “European Terrestrial Reference System 1989”.

iii) The CRS is based on a geodetic datum which is a member of the ETRS89 datum ensemble (EPSG datum code 6258).

   Software requiring to determine which geodetic CRSs are based on datums included in the ETRS89 ensemble should note that the codes for these datums are in the DatumEnsembleMember table.

6. How to merge data referenced to different realizations of ETRS89

The relationships between realizations of ETRS89 in the EPSG Dataset are shown diagrammatically in figure 1. They fall into five categories:

i) Direct transformations between national realizations.

ii) The ETRS89 datum ensemble.

iii) Transformations that may be used to form an operation pipeline:

   a. Between national realizations and an ETRFyy.

   b. Between an ETRFyy and the hub CRS, ETRF2000.

   c. Special provisions for Nordic and Baltic countries.

Collectively these allow data sets referenced to different realizations to be merged together.

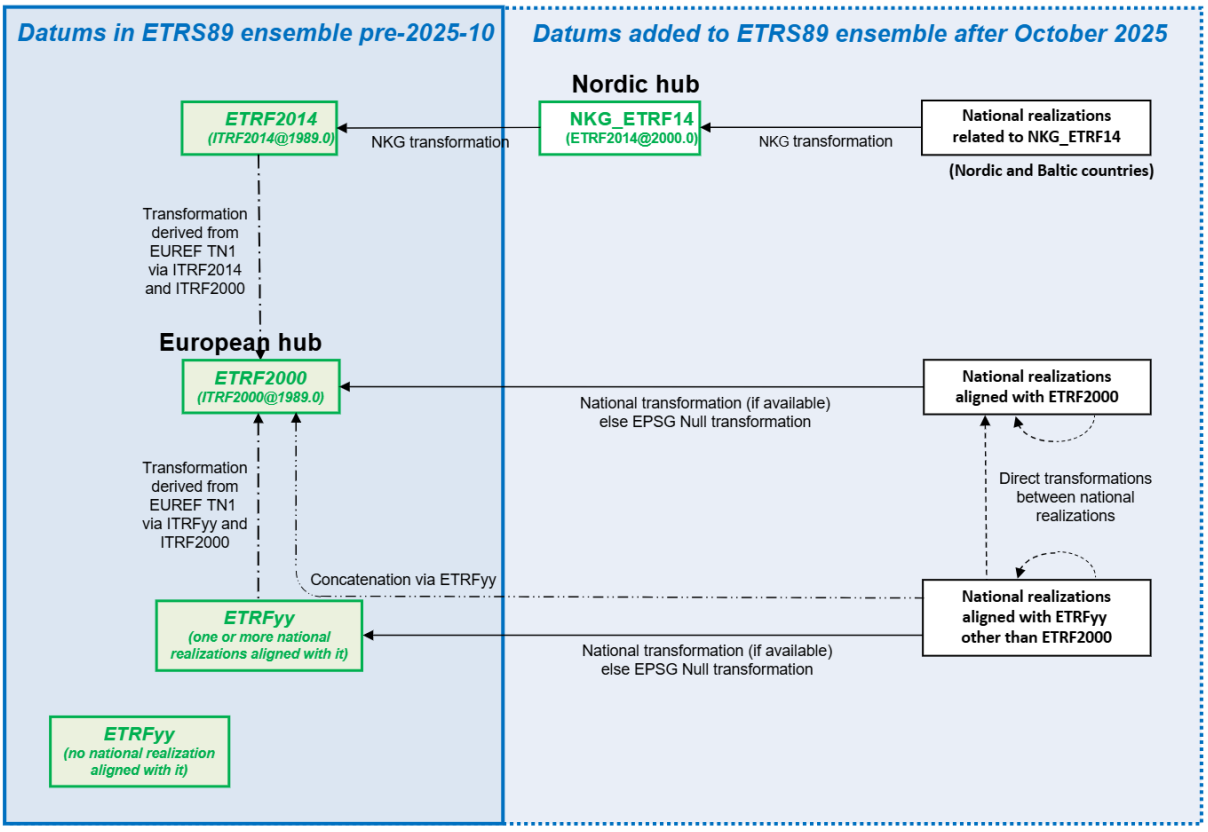


Figure 1: Relationships between realizations of ETRS89 in the EPSG Dataset.

Users requiring to merge spatial data referenced to different realizations of ETRS89 potentially have several routes to do this:

1) Using a direct transformation.

2) Utilizing the ETRS89 datum ensemble.

3) Using a pipeline operation through a hub.

6.1. Direct transformations between national realizations

Following standard coordinate transformation best practice, any documented direct transformation between the realizations (short dashed lines in Figure 1) should always be used in preference to alternatives. For example, to transform data from ETRS89-ITA [IGM95] to ETRS89-ITA [RDN2008], the IGM transformation (EPSG coordinate operation code 9737) should be used. Conversely, if a direct transformation is available, routes through a hub should not be used.

6.2. Utilizing the ETRS89 datum ensemble for transformations

The datum ensemble may be utilized in transformations in two very different ways:

1) If there is a requirement to transform data to or from a CRS based on the ETRS89 datum ensemble (EPSG:4936, EPSG:4937 or EPSG:4258), transformations applicable to an explicit national realization may also be utilized.

2) If there is a requirement to merge datasets referenced to CRSs both of which have datums that are members of the datum ensemble, and lower accuracy is acceptable, a direct merger avoiding transformation may be possible.

These are discussed in the two subsections below.

A difficulty with using datum ensembles is that their use requires data mining. In some circumstances, using name and/or alias may produce results more quickly, but because of issues with uncertainty in data population and text matching of name, these results will be less complete and less reliable than from using the datum ensemble.

6.2.1. Supplementing transformations using the ETRS89 ensemble

Transformations and concatenated operations that are associated with a national CRS that is a member of the ETRS89 datum ensemble may of course be applied directly to coordinates referenced to that CRS as in 6.1 above, but in addition may be applied to an equivalent CRS whose datum is the datum ensemble (EPSG:4936, EPSG:4937 or EPSG:4258). Conversely, transformations applicable to an ensemble-based CRS (EPSG:4258) are (i) those directly documented, and (ii) other transformations from CRSs whose datum is a member of the ensemble. For example:

- Assume two documented transformations applicable to Austria from MGI to ensemble-based ETRS89 (EPSG:4258) are variants (1) and (5).
- CRS ETRS89-AUT [2002] applicable to Austria is based on a datum which is a member of the ETRS89 datum ensemble. It is associated with a transformation MGI to ETRS89-AUT [2002] variant (8). This transformation may also be applied as a transformation from MGI to ensemble-based ETRS89.

The reverse of this is not true – transformations associated with CRSs based on an ensemble should not be applied to specific realizations. This is an inherent limitation of spatial data being referenced to the ensemble: its association with a specific realization is simply not specified.

- Therefore three transformations are available for transforming between MGI and ensemble-based ETRS89 (EPSG:4258). But only one of these, variant 8, is available for transforming between MGI and the Austrian national realization ETRS89-AUT [2002].

The software logic for finding a transformation between two specified CRSs where one has a datum ensemble is:

i) list all transformations that are an exact match of source and target CRS, or have source and target transposed and the CT method is reversible, and

ii) find additional transformations that may also be used by using the datum ensemble.

   a) Check whether the source or target CRS is based on a datum that has the conventional RS set (and therefore is a member of a datum ensemble);

   b) Find transformations associated with CRSs based on members of the datum ensemble and the second of the two specified CRSs;

   c) Test whether that datum ensemble is the same as that for the ensemble-based specified CRS;

   d) Test that the extent and accuracy of the transformation meet requirements.

6.2.2. Merging datasets using the datum ensemble directly

If no direct transformation is available *and the highest accuracy is not required*, then as long as either the CRSs for both realizations have a datum that is a member the ETRS89 ensemble, or one of the two CRSs itself is based on the ETRS89 datum ensemble, the spatial data sets may be merged without application of a transformation. Applying a null transformation will have the same effect. (Coordinate conversions to ensure commonality of map projection, axis direction, units, etc. may be required). The ETRS89 datum ensemble is described in section 2 and shown diagrammatically in Figure 1.

The accuracy at which the assumption of no significant difference between realizations applies is given by the *datum ensemble accuracy* attribute. For the ETRS89 datum ensemble, this has been recorded in the EPSG Dataset as 0.1 m.

Examples of CRSs which may be related using the ETRS89 datum ensemble are:

- ETRS89-PRT [1995] and ETRS89-ESP [REGENTE] in the geographic 3D domain.
- ETRS89-NOR [EUREF89] and ETRS89 in the geographic 2D domain (where ETRS89 is ensemble-based).

In effect, the first example implies that the merged dataset is referenced to the ETRS89 ensemble (CRS 4937).

The second of these examples may be particularly relevant for maritime countries where offshore spatial data referenced to the ETRS89 ensemble-based CRS is to be merged with onshore data referenced to a national realization, or to spatial data that was referenced to the ETRS89 ensemble before the national realization was documented in the EPSG Dataset.

6.3. Operation pipelines through a hub

6.3.1. Relationship of a national realization to ETRFyy

EPSG Dataset policy is to include a transformation between a national realization of ETRS89 and the ETRFyy to which the realization is aligned. If a national authority publishes one, this will be used. When no information is available, an EPSG null transformation is created. An EPSG null transformation is one created artificially by EPSG equating the coordinates of a location referenced to different CRSs. It ignores rigorous geodesy. Typically, it has been defined using the Geocentric translations method where all parameter values are zero, i.e., tX=tY=tZ=0.

At a location, coordinates of CORS stations referenced to the national realization may differ—at the millimetre or sub-millimetre level—from coordinates referenced to the EPN in a specified ETRFyy because the national network adjustment usually includes stations which are not part of the EPN, causing small differences in the adjusted networks. But for most practical purposes, equality can be assumed. However, this may be valid only at the epoch at which the national realization adjustment was made. At times other than the datum anchor epoch, ground movement caused by for example glacial isostatic adjustment (GIA) or tectonic plate deformation, the national realization may diverge from its ETRFyy.

To allow for possible crustal deformation over the life of the source and target CRSs, EPSG assigns to null transformations between the national realization and its ETRFyy a default coordinate operation accuracy of 0.1m. This value is the same as the ETRS89 ensemble accuracy. In areas with little deformation this operation accuracy value is overly conservative. Therefore, EPSG null transformations include standard remarks "*Accuracy is approximately 1 cm at the [name of national realization] anchor epoch of [epoch] but deteriorates with time due to tectonic deformation and/or glacial isostatic adjustment*". One centimetre at epoch of observation is the criteria for meeting EUREF class B alignment to ETRS89 [EUREF, 2021].

6.3.2. Choice of hub

In the EPSG Dataset the chosen hub for merging data referenced to realizations of ETRS89 is ETRF2000, as, following EUREF recommendation [EUREF, 2010], most countries have aligned their national realization to ETRF2000.

Should EUREF update its recommendation for national realizations to be aligned with ETRF2000 to a later pan-European realization of ETRS89, additional transformations to this new hub will be added to the EPSG Dataset.

6.3.3. Relationship between pan-European realizations of ETRS89

The transformation from ETRFyy to ETRF2000 may be derived through adding together three time-dependent Helmert transformations defined in EUREF Technical Note 1 and IERS technical notes:

i) ETRFyy to ITRFyy

ii) ITRFyy to ITRF2000

iii) ITRF2000 to ETRF2000

Before they are concatenated, these three transformations need to be reduced to the same parameter reference epoch. The concatenation is a time-dependent transformation. For ETRFyy's to which one or

more national realizations are aligned, IOGP has derived a single transformation from ETRFyy to ETRF2000 by using this concatenation, shown in Figure 1 with a dash-dot (dash-single dot) line.

For countries which did not align their national realization to ETRF2000, the national realization of ETRS89 may be related to ETRF2000 by concatenating (i) the transformation from national realization to ETRFyy with (ii) the transformation from ETRFyy to ETRF2000, as shown in Figure 1 with a chained (dash-dot-dot) line.

6.3.4. Relationship between national realizations using an operation pipeline through the hub

The EPSG Dataset includes transformations which permit concatenation through a hub when no direct transformation is available. For merging spatial data referenced to different national realizations of ETRS89, for example Spain and Portugal, ETRF2000 is used as the hub. The operation pipeline through this is (i) national realization 1 to ETRF2000 concatenated with (ii) ETRF2000 to national realization 2.

This concatenation is analogous to that using WGS 84 described in EPSG Guidance Note 7-5 [IOGP, 2022]. Note that because the transformations involved have lower (better) accuracy, using ETRF2000 as the hub will induce significantly less error in the target coordinates than using WGS 84 as the hub.

Whether merging data sets through an operation pipeline is appropriate requires careful consideration, as the transformations involved will usually have been derived from coordinates within one of the countries and therefore apply only to part of the total area. In principle a transformation is applicable only within its stated extent.

6.3.5. Nordic and Baltic countries

The national realizations of the Nordic and Baltic countries are treated differently to the relationships described in the previous subsections. The transformations involved include modelling of time-dependent coordinate changes due to post-glacial isostatic adjustment. The Nordic Geodetic Commission (NKG) defines a hub CRS—currently NKG_ETRF14 (EPSG CRS code 10805), a static snapshot of ETRF2014 at epoch 2000.0—to which each Nordic and Baltic national realization is related.

For merging spatial data referenced to different national realizations of the Nordic and Baltic countries, for example spatial data from Norway and Sweden, NKG_ETRF14 is used as the hub. The operation pipeline is (i) national realization 1 to NKG_ETRF14 concatenated with (ii) NKG_ETRF14 to national realization 2.

For merging spatial data referenced to a national realization of one of the Nordic and Baltic countries with data reference to a national realization outside of that area, for example Lithuania and Poland, ETRF2000 is used as the hub. The operation pipeline is (i) Nordic national realization to NKG_ETRF14 concatenated with (ii) NKG_ETRF14 to ETRF2000 and then with (iii) ETRF2000 to non-Nordic national realization.

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International Association of Oil & Gas Producers

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The complete texts may be found on the EPSG Registry web site under Support Documentation.

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